Teaching Interests and Philosophy

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Teaching philosophy: fundamentals, experiments, and accountable decisions

I want students to leave an ECE classroom able to explain a principle, implement it, and recognize when the result should not be trusted. My teaching philosophy follows a recurring engineering cycle: establish a mathematical or physical model, make a prediction, implement a circuit, program, or simulation, compare the result with measurement, and explain the discrepancy. That cycle connects theoretical rigor with intellectual curiosity and turns a "correct answer" into a reasoned design decision. It is also an inclusive point of entry: students whose interests lie in electronics, communications, robotics, or bioengineering can contribute different strengths to the same technical investigation.

I have taught undergraduate Computer Programming, Communication Systems, and Wireless Communication at the University of Lahore; contributed to communications teaching and laboratory sessions at the University of Engineering and Technology, Taxila; and supported Bachelor's and Master's research and thesis work at COMSATS University Islamabad. These roles taught me the importance of sequencing concepts carefully and giving constructive feedback before the final assessment. My research experience further sharpened how I teach the gap between nominal models and real conditions. For example, my dissertation found that seemingly modest measurement and positioning errors can change optimization outcomes; this is a tangible lesson in why engineering assumptions must be tested, not merely stated.

Teaching contributions to Lafayette's curriculum

My immediate teaching strengths align with Signals and Systems (ECE 331), Communication Systems (ECE 332), and programming-based engineering problem solving. My experience in signal processing, wireless simulation, Python, statistical modeling, and research mentoring would also support Digital Signal Processing (ECE 434), Communications Networks (ECE 436), and upper-level projects involving acquisition and analysis of experimental signals. I would design applied electives such as Reliable AI for Networked Systems or Wearable Sensing and Cyber-Physical Systems, scaled to Lafayette's elective and interdisciplinary interests.

I recognize that the advertised search specifically emphasizes electronics, instrumentation, computer architecture, cyber-physical systems, and embedded systems. Rather than present communications teaching as equivalent to formal experience teaching those core courses, I would contribute in concrete, relevant ways. My work on wearable inertial sensing provides an applied context for Sensors and Electronic Systems (ECE 427); my error-aware optimization and feedback research provides examples for cyber-physical systems and control; and my programming experience can support sensor-to-software integration and project modules in Embedded Systems (ECE 414). Teaching the broader electronics and hardware-design curriculum would require deliberate preparation, coordination with established laboratory faculty, and attention to Lafayette's particular learning outcomes and equipment.

How I would teach a laboratory or design sequence

Consider a short sequence on sensor measurement and robust decision-making. Students first predict sampling and noise effects from a mathematical model. Next, they inspect or acquire accelerometer data, select a sampling rate, and implement a simple filter. They then compare a threshold-based detector with a small learning model and deliberately test both under changed sensor orientation, missing samples, and shifted operating conditions. The assessment is not just whether the classifier works on one test set, but whether the team explains failures, documents assumptions, and justifies a design change. This sequence connects ECE signals and programming to instrumentation, cyber-physical behavior, and bioengineering without assuming an expensive research platform.

I would scaffold projects through milestones: problem statement, baseline implementation, calibration and test plan, design review, and reflective report. When students encounter unexpected results, I would ask them to distinguish bugs, measurement artifacts, insufficient modeling assumptions, and genuine model failure. My dissertation work on cell-outage compensation offers another instructional case: students can compare a rule-

based policy and a simple learning-based controller while examining the trade-off between recovering service to affected users and protecting users already served. Students would be expected to communicate the trade-off clearly, not simply report the highest aggregate score.

Assessment, access, and inclusive learning

I assess engineering reasoning explicitly. In addition to exams and problem sets, lab rubrics should separately evaluate the model, implementation, test quality, interpretation, documentation, and team contribution. Early low-stakes concept checks reveal where students need support; formative feedback on proposals and prototypes makes improvement possible before a final demonstration. I would use concise written guidance, accessible examples, office hours, and structured team roles so that students with uneven prior coding or hardware experience can participate fully. Rotating responsibilities for coding, measurement, testing, and presentation can prevent one student from becoming the permanent technical lead.

Inclusiveness also means making the path to a high standard visible. I would explain prerequisites and unfamiliar terminology, provide a reproducible starting template without removing meaningful decisions, and support multiple ways of showing understanding in preparation for a final shared technical standard. I would seek student feedback and draw on Lafayette's Hanson Center for Inclusive STEM Education and Center for the Integration of Teaching, Learning, and Scholarship to improve my methods. I would not substitute presumed uniform preparation for evidence gathered from the actual class.

Mentoring undergraduates through authentic research

My aim is to make undergraduate research an extension of disciplined learning rather than a privilege reserved for students with advanced ML backgrounds. A student may begin by reproducing a baseline from my dissertation, documenting a simulator, or quantifying one failure mode in sensor data. More experienced students can develop an experiment, benchmark a proposed improvement, and prepare a poster or manuscript where the results warrant it. I would meet regularly with teams, establish a shared code and lab notebook, clarify authorship expectations, and encourage students to explain both positive and negative findings. The result should be students who understand their own evidence and can defend the limits of their claims.

Lafayette's small-class environment offers a particularly meaningful setting for this kind of mentorship. My goal is for students to become confident engineers who can connect foundations to hardware or software, ask responsible questions about system reliability, and communicate their choices to both technical and nontechnical audiences.